



## Article

# Solar Flare Forecast: A Comparative Analysis of Machine Learning Algorithms for Predicting Solar Flare Classes

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## Abstract

Solar flares are among the most powerful and dynamic events in the solar system, resulting from the sudden release of magnetic energy stored in the Sun's atmosphere. These energetic bursts of electromagnetic radiation can release up to  $10^{32}$  erg of energy, impacting space weather and posing risks to technological infrastructure and therefore require accurate forecasting of solar flare occurrences and intensities. This study evaluates the predictive performance of three machine learning algorithms—Random Forest (RF), k-Nearest Neighbors (kNN), and Extreme Gradient Boosting (XGBoost)—for classifying solar flares into four categories (B, C, M, X). Using 13 parameters of the SHARP dataset, the effectiveness of the models was evaluated in binary and multiclass classification tasks. The analysis utilized 8 principal components (PCs), capturing 95% of data variance, and 100 PCs, capturing 97.5% of variance. Our approach uniquely combines binary and multiclass classification with different levels of dimensionality reduction, an innovative methodology not previously explored in the context of solar flare prediction. Employing a 10-fold stratified cross-validation and grid search for hyperparameter tuning ensured robust model evaluation. Our findings indicate that RF and XGBoost consistently demonstrate strong performance across all metrics, benefiting significantly from increased dimensionality. The insights of this study enhance future research by optimizing dimensionality reduction techniques and informing model selection for astrophysical tasks. By integrating this newly acquired knowledge into future research, more accurate space weather forecasting systems can be developed, along with a deeper understanding of solar physics.



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**Keywords:** solar flares; machine learning; space weather forecast

## 1. Introduction

Solar flares are among the most powerful and dynamic events in the solar system, resulting from the sudden release of energy of the Sun's atmosphere [1] (Code available at <https://github.com/juliabringewald/Solar-Flare-Forecast> (accessed on 11 March 2025)).

These energetic bursts of electromagnetic radiation originate from stressed magnetic fields in active regions (ARs). The impact of solar flares extends beyond the Sun, influencing space weather and posing risks to technological infrastructure such as satellite communications, power grids, and aviation systems [2,3]. Given their potential for disruption depending on their intensity, presented in Table 1, accurate forecasting of solar flare occurrences and intensities is essential for space weather prediction and planning satellite operations.

As the closest star to Earth, the Sun plays a crucial role in shaping the space environment, with solar activity driving significant variations in space weather conditions. X-ray flux, a key indicator of solar activity, represents high-energy emissions from solar flares and other dynamic processes on the Sun's surface. Accurate prediction of variations is critical for safeguarding space missions, satellite operations, and ground-based technological infrastructure. Solar flares, alongside coronal mass ejections (CMEs), can trigger geomagnetic and particle disturbances, leading to technological and societal consequences [4]. Observational and theoretical studies indicate that solar flares and CMEs are powered by the rapid release of magnetic free energy stored in the corona, facilitated by magnetic reconnection [5]. The build-up of this energy is caused by the evolution of the magnetic field on the photosphere. Although direct measurements of the coronal magnetic field remain challenging, observations of the photospheric magnetic field provide critical insights into energy accumulation and flare-triggering mechanisms [6]. Various parameters, including sunspot classification, vertical electric currents, magnetic free energy, shear, and helicity, characterize ARs and serve as indicators for solar flare prediction.

Despite significant research efforts, the connection between AR magnetic complexity and flare productivity remains only partially understood. However, the fundamental coupling between the photosphere and the corona has motivated the use of photospheric magnetic field parameters in flare prediction models, particularly through statistical and Machine Learning techniques [7,8]. Unlike traditional statistical methods that require assumptions about data distribution, Machine Learning algorithms can learn the inherent patterns in data, which presents a beneficial approach for predicting space weather. Many previous studies have relied on parameters derived from the line-of-sight (LOS) component of the photospheric magnetic field to estimate the probability of flare occurrence as well as the intensity [8,9].

The SHARP dataset (Space Weather HMI Active Region Patch), derived from NASA's Solar Dynamics Observatory (SDO) Helioseismic and Magnetic Imager (HMI), provides continuous high-cadence measurements of the Sun's vector magnetic field. Vector magnetic data provides a more comprehensive view than scalar data by including the direction (inclination and declination) of the magnetic field.

This data enables the computation of key magnetic parameters used in flare forecasting. This study utilized SDO/HMI data to predict the maximum magnitude of solar flares classified by GOES classes (B, C, M, X). It should be emphasized that the data set utilized in this research is publicly accessible. Dataset: (Liu et al. [10]).

**Table 1.** Solar flare intensity and associated effects.

| Solar Flare Class | Associated X-Ray Flux $I$ (W/m <sup>2</sup> ) | Possible Effects on Earth                                                     |
|-------------------|-----------------------------------------------|-------------------------------------------------------------------------------|
| B                 | $I < 1 \times 10^{-6}$                        | None                                                                          |
| C                 | $1 \times 10^{-6} \leq I < 1 \times 10^{-5}$  | Possible effects on space missions                                            |
| M                 | $1 \times 10^{-5} \leq I < 1 \times 10^{-4}$  | Radio transmission blackouts; potential harm to astronauts outside spacecraft |
| X                 | $I \geq 1 \times 10^{-4}$                     | Damage to satellites, communication systems, power grids, and electronics     |

Recent studies have demonstrated the effectiveness of Machine Learning in solar flare forecasting using photospheric magnetic field data. For instance, a Machine Learning algorithm was developed using SDO/HMI vector magnetic field data to predict flare occurrence, establishing a benchmark for physics-informed classification models [11]. More recently, a comparative analysis evaluated multiple Machine Learning models—including Random Forest (RF) and Extreme Gradient Boosting (XGBoost)—on the same SDO/HMI dataset, reinforcing the relevance of these algorithms for flare prediction tasks [12]. Additionally, former studies applied advanced deep learning architectures such as LSTM and Decomposition-LSTM to flare forecasting, showcasing the potential of temporal pattern recognition techniques in space weather modeling [13]. While these studies differ in scope, methodology, and target variables, they collectively underscore the growing role of Machine Learning in solar physics.

This study employed three Machine Learning models—RF, k-Nearest Neighbors (kNN), and XGBoost—to evaluate solar flare classification performance into GOES categories.

The selection of these specific models is based on their previously studied ability regarding classification tasks within the field of astrophysics. >RF have been noted for their strong ability to rank the relative importance of features [14], while XGBoost is widely regarded for its consistent and robust performance across a range of classification tasks [15]. In addition, the k-Nearest Neighbours (kNN) classifier has repeatedly demonstrated high levels of accuracy, supporting its reputation as a dependable tool for classification studies [16].

This research aims to introduce a new layer of comparison within the three primary Machine Learning models—kNN, RF, and XGBoost. The performance of these models is evaluated in both binary and multiclass classification tasks. Additionally, the impact of dimensionality reduction is assessed by comparing model performance using 8 PCs versus 100 PCs derived from Principal Component Analysis (PCA), capturing different amounts of data variance.

Unlike previous works, our approach is the first, to our knowledge, to compare these models across both classification types and different levels of dimensionality reduction in the context of solar flare prediction. This dual-layered comparison offers new insights into model behavior under varying data representations and classification schemes, addressing a gap that has not yet been explored in this field.

The findings of this study contribute to the advancement of space weather prediction, emphasizing the potential of Machine Learning-driven techniques to improve prediction systems for solar flares. By improving the accuracy and reliability of forecasts, such techniques could reduce the impact of severe space weather events on technological systems, while also deepening our understanding of solar physics.

## 2. Dataset

For this study, a pre-processed dataset of solar flares based on Space-weather HMI Active Region Patches (SHARP) data, originally created by Liu et al. [10], was utilized. The SHARP dataset, developed by the SDO/HMI team Bobra et al. [11], provides magnetic field measurements and derived parameters for automatically identified and tracked active regions (ARs). Additionally, the cgem. Lorentz data series provides estimations of Lorentz forces to help analyze AR dynamics [17].

Liu et al. [10] constructed their dataset by selecting ARs that produced flares over a 6.5-year period (May 2010–December 2016), covering the peak of Solar Cycle 24. Flare records were obtained from the GOES X-ray flare catalogs provided by the National Centers for Environmental Information (NCEI). These records were merged with USAF Solar Observing Optical Network H-alpha flare listings [10], ensuring comprehensive flare identification and classification.

The dataset follows the GOES flare classification (B, C, M, X), as shown in Table 1, where each AR is categorized by the highest GOES-class flare it produced [18,19]. To ensure data quality, only C-, M-, and X-class ARs with flares occurring within  $\pm 70^\circ$  of the central meridian were included, while B-class ARs were manually verified through solarmonitor.org. Furthermore, the dataset includes 13 key magnetic field parameters, presented in Table 2, identified as strong predictors of solar flare activity [11].

**Table 2.** Overview of SDO/HMI Magnetic Parameters.

| Parameter | Description                                        |
|-----------|----------------------------------------------------|
| ABSNJZH   | Absolute value of net current helicity             |
| AREA_ACR  | Area of strong field pixels in active region (AR)  |
| EPSZ      | Sum of z-component of normalized Lorentz force     |
| MEANPOT   | Mean photospheric magnetic free energy             |
| R_VALUE   | Sum of flux near polarity inversion line           |
| SAVNCPP   | Sum of the modulus of the net current per polarity |
| SHRGT45   | Fraction of area with shear $>45^\circ$            |
| TOTBSQ    | Total magnitude of Lorentz force                   |
| TOTFZ     | Sum of z-component of Lorentz force                |
| TOTFOT    | Total photospheric magnetic free energy density    |
| TOTUSJH   | Total unsigned current helicity                    |
| TOTUSJZ   | Total unsigned vertical current                    |
| USFLUX    | Total unsigned flux                                |

### 3. Methodology

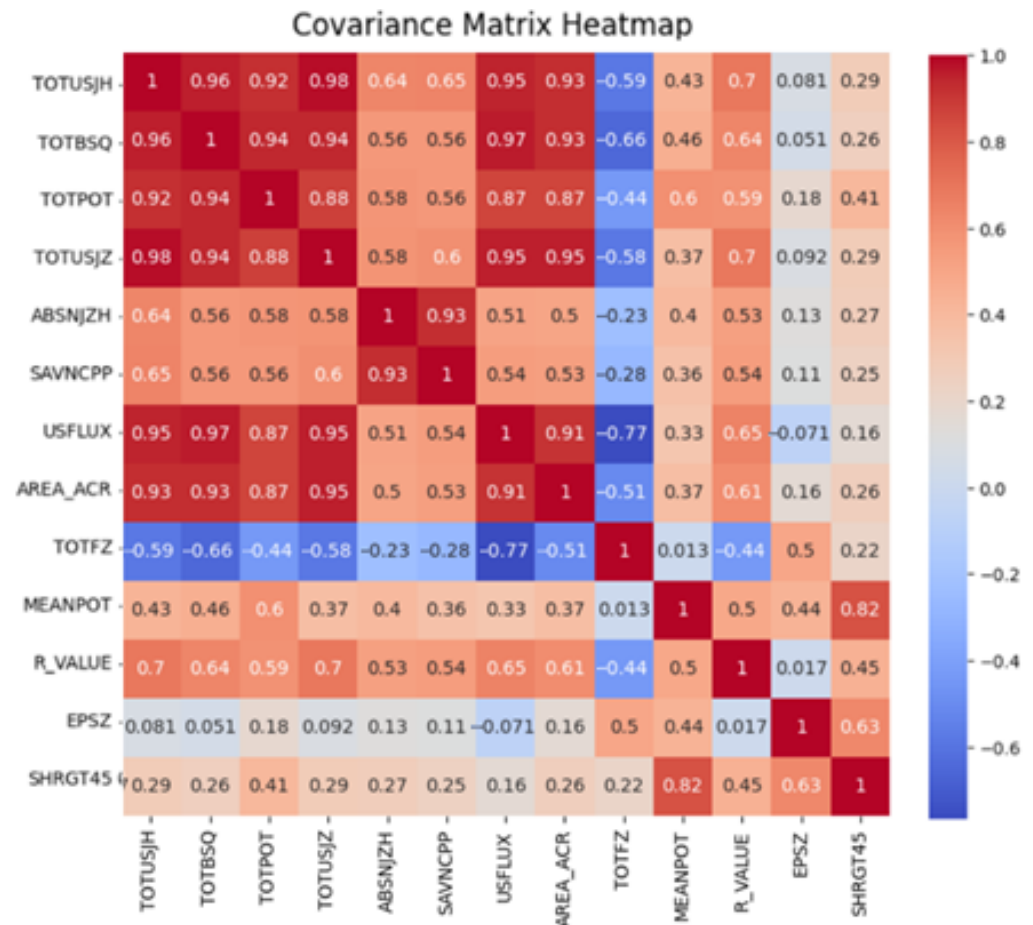
#### 3.1. Experimental Setup

The dataset consists of solar activity parameters, including sunspot counts, magnetic field measurements, and previous flare occurrences. Several Machine Learning models, including logistic regression, decision trees, and neural networks, are trained and evaluated based on their predictive accuracy.

#### Feature Engineering

PCA method is used for reduction of dimensionality of the parameter space or as to say the number of variables. The 13 magnetic parameters are standardized using StandardScaler, because PCA relies on the data being normally distributed with each feature having a similar range. After standardizing the data, interaction features of the standardized data are calculated, visualized in Figure 1. Interaction features can help capture non-linear relationships between the original features that may not be apparent when considering them individually. Also, this approach helps to reduce overfitting by capturing the underlying structure of the data more effectively.

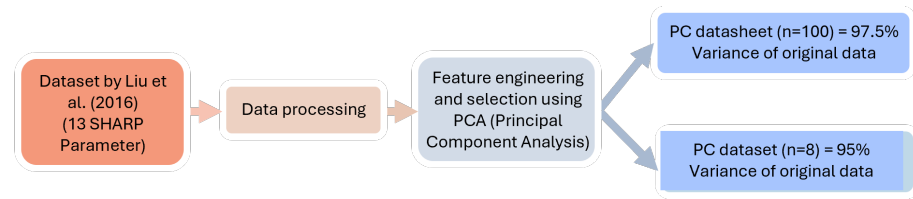
How many of the interaction features are most beneficial for the model quality is determined by a certain variance threshold of the data the features should capture. Capturing as much variance as possible refers to preserving the diversity of the data, while reducing its overall dimensionality. This is important to keep the full representation of data diversity while reducing your initial amount of features (in this case the interaction features). In simple terms, capturing a lot of variance in your dataset is like summarizing a book while keeping all the main ideas and discarding the unnecessary details.



**Figure 1.** Covariance heatmap of magnetic parameters as explained in Table 2. Note: A covariance heatmap is a visual representation of the covariance matrix of a dataset's features. Covariance measures the strength of joint variability between two or more variables, as to say how much two variables change together. Positive covariance indicated that as one variable increases, the other tends to increase as well. Negative covariance indicates that as one variable increases, the other tends to decrease. If two parameters are statistically independent, their covariance should be close to zero, indicating little to no linear relationship.

In relation to covariance calculation, PCA derives the features with the strongest covariance of interaction features, while capturing a determined amount of variance from the original data.

In this study, a threshold of 95% cumulative explained variance was used to determine the number of PCs required to retain most of the information in the dataset. This means that the first 8 PCs were sufficient to capture 95% of the variability present in the original interaction feature set. This refers to the total amount of information or variability captured by a certain number of principal components in the interaction feature dataset. To observe the impact of capturing more variance, and therefore more original information from the original dataset, the use of 100 components was analyzed, capturing 97.5% of the variance. This process is visualized in Figure 2. The analysis did not include more than 100 components due to limitations in computational resources.



**Figure 2.** Visualization of the feature engineering process. Reference: [10]

This approach examines how using different amounts of original data and the risk of overfitting from too-detailed data affect model performance. These components (PC = 8 and PC = 100) are selected and retained in two different data sets, which will be further used in the Machine Learning process.

### 3.2. Algorithms

#### 3.2.1. kNN

The kNN algorithm is a supervised learning classifier, introduced by Cover and Hart in 1967 [20]. It operates on the principle of proximity, predicting the output based on the most similar data points in the feature space. For classification, kNN assigns a class label to a new sample based on the majority vote among its K-nearest neighbours, where K represents the number of neighbours considered. This can be more accurately described as “plurality voting” in multi-class settings.

The algorithm’s simplicity and intuitive approach make it popular and widely applied across various fields. kNN requires recalculating the distance between the input and all data points whenever new data is introduced, which can be computationally intensive compared to other models.

Key hyperparameters of kNN include the number of nearest neighbours (K), the weighting of each neighbour (whether all neighbours contribute equally or closer neighbours have a larger weight), and the distance metric used for calculating proximity between samples.

#### 3.2.2. RF Classifier

The RF algorithm, trademarked by Leo Breiman and Adele Cutler [21], is a widely used Machine Learning method known for its flexibility and ease of use. It is an ensemble learning technique that combines the output of multiple decision trees to achieve a single result, making it suitable for both classification and regression tasks.

In the training phase, each decision tree in the forest is trained on a bootstrap sample of the original data, where approximately 37% of instances are duplicated. Additionally, at each node split, a random subset of the input features is selected to determine the best split. This dual randomization—bootstrap sampling and feature selection—helps reduce overfitting and increase the model’s robustness. For classification tasks, the final prediction is determined by majority voting among the trees.

#### 3.2.3. XGBoost

XGBoost is a distributed, open-source Machine Learning library that excels in speed, efficiency, and scalability for large datasets [22]. It employs gradient-boosted decision trees, a supervised learning algorithm utilizing gradient descent for optimization. XGBoost starts with a weak learner, known as the base learner, and builds new trees in an additive manner to correct the base learner’s errors. The algorithm calculates the residuals, which are the differences between the predicted and actual values, and uses these residuals to guide the creation of subsequent trees. The model’s performance is evaluated using loss functions, such as mean squared error for regression and cross-entropy loss for classification.



Gradient descent minimizes the loss, improving model performance with each iteration. XGBoost's ability to handle both classification and regression tasks, combined with its efficient handling of large datasets and robust performance, makes it a popular choice in Machine Learning applications.

In simpler words: XGBoost begins with a simple model that makes basic predictions. Then, it adds new models (decision trees) in a step-by-step process, each one learning from the mistakes of the previous models. These trees try to fix the errors made by earlier predictions. The predictions from all the trees are combined to make a final, more accurate prediction. The algorithm uses a technique called gradient descent to adjust and minimize the errors, making the model better with each new tree added.

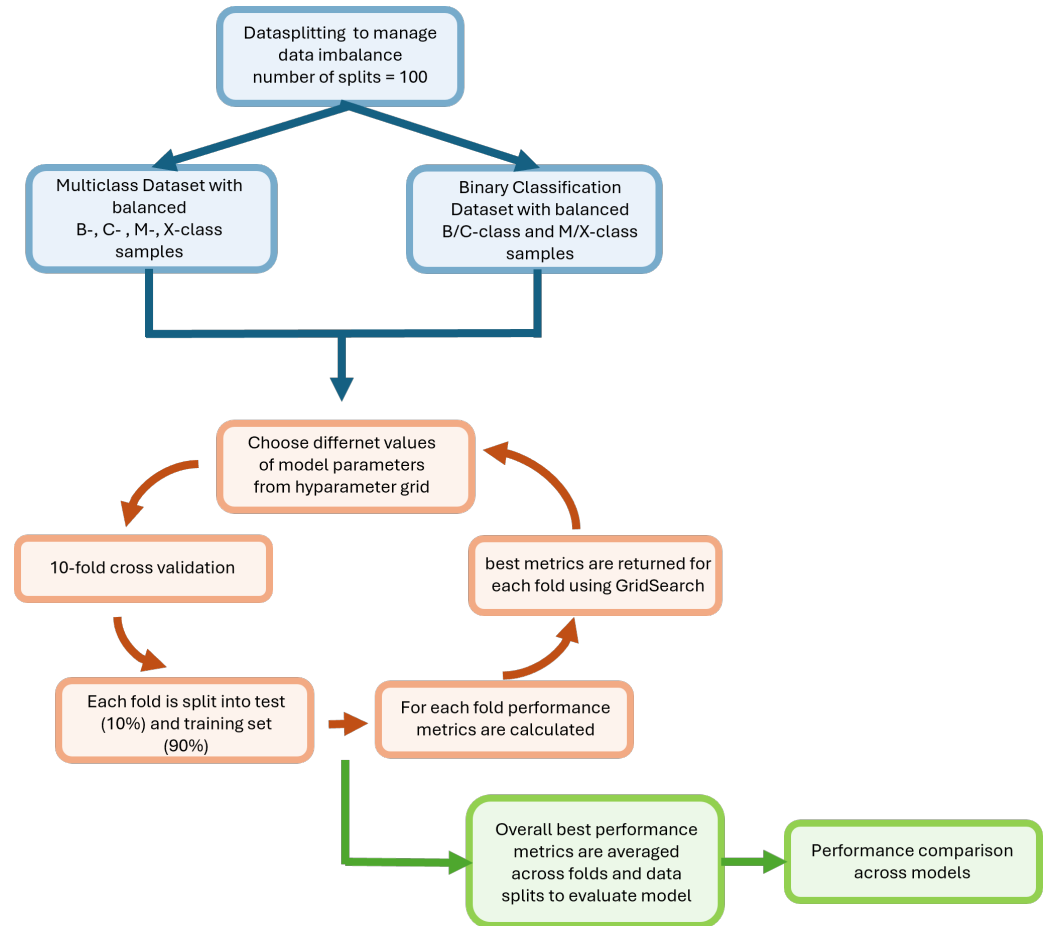
### 3.3. Training and Evaluation

For the flare prediction, every algorithm operates in classification mode. During the training phase, each input training sample includes interaction parameter values ( $n = 8$  or  $n = 100$  as described in Section 3.1) of an AR and its corresponding maximum GOES flare class, which represents the target variable. In the testing/validation phase, these parameters of an AR sample serve as the model input, and the classifier predicts the GOES class of this AR. The dataset utilized [10] exhibits an imbalance in the number of AR samples, with 552 C-class samples, 23 X-class samples, 142 M-class samples, and 128 B-class samples. To address this class imbalance, which poses a significant challenge in Machine Learning [23], the approach suggested by Liu et al. [10] was adopted. From the total 552 C-class samples, 142 unique samples were randomly selected to create a balanced dataset. To ensure robustness and avoid bias, this random selection was repeated 100 times, producing 100 datasets for multi-class classification. Each dataset contains 128 B-class, 142 C-class, 142 M-class, and 23 X-class AR samples, as illustrated in the models' training method in Figure 3.

Additionally, for binary classification, another set of 100 datasets was constructed to facilitate comparisons with previous work. The B- and C-class samples were combined to form the B/C class, while the M- and X-class samples were combined to form the M/X class. For each of the 100 datasets, 165 unique B/C-class AR samples were randomly selected and paired with 165 M/X-class samples, ensuring balanced binary classification datasets. This data splitting approach prevents misclassifications by Machine Learning models due to the previous class imbalance.

Given that M/X-class solar flares produce the most significant effects on Earth, as presented in Table 1, adopting a binary classification approach is beneficial for predicting the most intense solar flares. This method improves the prediction of solar flares that have substantial terrestrial impacts.

To evaluate the performance of various Machine Learning algorithms, the widely used 10-fold cross-validation (CV) method is applied. For each dataset, stratified 10-fold partitioning is performed using `StratifiedKFold()`, ensuring that each fold contains nearly equal-sized groups with balanced class distributions. The algorithm is then trained using nine folds of the data, while the remaining fold is used for validation. This process is repeated for each combination of hyperparameters being evaluated using the `GridSearch()` function. Hyperparameters are settings that control the learning process and are not learned from the data. Proper tuning helps improve model accuracy, reduce overfitting, and ensure that the model generalizes well to new and unseen data.



**Figure 3.** Visualization of models' training and evaluation method.

During the cross-validation, the model's performance is assessed using predefined metrics such as accuracy, ROC AUC, F1 score, and PR AUC. The indepth functioning of the performance metrics is described as following.

$$\text{Overall Accuracy} = \frac{TP + TN}{TN + FP + FN + TP}$$

$$\text{F1 score} = \frac{2 \times TP}{2 \times TP + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{PR AUC} = \int_0^1 \text{Precision } d(\text{Recall})$$

$$\text{True Positive Rate (TPR)} = \frac{TP}{TP + FN}$$

$$\text{False Positive Rate (FPR)} = \frac{FP}{FP + TN}$$

$$\text{ROC AUC} = \int_0^1 \text{TPR } d(\text{FPR})$$

1. True positives (TP): number of samples correctly predicted as "positive".
2. False positives (FP): number of samples wrongly predicted as "positive".
3. True negatives (TN): number of samples correctly predicted as "negative".



4. False negatives (FN): number of samples wrongly predicted as “negative”.

The performance metrics from the 10 folds are averaged to obtain the mean performance metric for each hyperparameter combination. This averaging process mitigates the impact of data variability and provides a robust evaluation.

The hyperparameter combination that yields the highest mean performance metric across the 10 folds is selected as the best-performing set of hyperparameters. Various standard performance metrics, such as Receiver Operator Curve (ROC AUC), accuracy, and precision–recall AUC and F1 are calculated to assess model performance, with AUC signifying Area Under the Curve and with values closer to 1 indicating better performance.

### Hyperparameters

Three algorithms were investigated in this study, and the details of their hyperparameter search grids are listed in Table 3.

**Table 3.** Hyperparameter search grid for kNN, RF, and XGBoost.

| Algorithm | Tuning Parameter                           | Search Grid                |
|-----------|--------------------------------------------|----------------------------|
| kNN       | Number of Nearest Neighbours (n_neighbors) | [3, 5, 7, 9]               |
| kNN       | Weight Metric                              | ['uniform', 'distance']    |
| kNN       | Distance Metric                            | ['euclidean', 'manhattan'] |
| RF        | Number of Features                         | [8, 9, 10, 11, 12, 13]     |
| RF        | Tree Depth (min_samples_split)             | [8, 10]                    |
| XGBoost   | Learning Rate                              | [0.1, 0.9]                 |
| XGBoost   | Subsample                                  | [0.01, 1]                  |

### Notes on Hyperparameters per model:

#### kNN

- Number of Nearest Neighbours: This parameter determines how many neighbours are considered when making a prediction. The choice of the number of neighbors directly affects the model’s bias–variance trade-off. A smaller number of neighbors (e.g., 3) can result in a model that captures local patterns well but might be sensitive to noise (low bias, high variance). On the other hand, a larger number of neighbors (e.g., 10) can smooth out predictions, making the model more robust to noise but potentially less sensitive to local variations (high bias, low variance).
- Metric of weights: This parameter decides how the distance between points affects their influence on the prediction. Using ‘uniform’ weights means all neighbors contribute equally, which is suitable when the data is uniformly distributed. ‘Distance’ weighting allows closer neighbors to have a greater influence, which can be beneficial when the relevance of neighbors decreases with distance.
- Metric of distance: This parameter defines how the distance between points is calculated. ‘Euclidean’ distance is the most commonly used metric and is appropriate when the feature space is continuous. ‘Manhattan’ distance is useful when dealing with high-dimensional data or when features are on different scales.

#### RF

- Number of features: This parameter specifies how many features are considered when splitting a node. More features can lead to a more complex model, while fewer features can simplify it.
- Depth of trees: This parameter controls the maximum depth of each tree in the forest. Deeper trees can capture more complex patterns but may also overfit the data. A depth

of 10 for example allows the model to capture complex patterns without becoming too complex and overfitting.

- Minimum number of sample splits: This parameter determines the minimum number of samples required to split an internal node. Setting this parameter to a higher value (e.g., 5) ensures that splits are only made when there is enough data to justify the split, preventing overfitting and improving the model's generalization ability.

*XGBoost*

- Learning rate: This parameter controls the step size at each iteration while moving toward a minimum of the loss function. A smaller learning rate can lead to more accurate models but requires more iterations.
- Subsample: This parameter specifies the fraction of samples used for fitting individual base learners. Lower values can prevent overfitting by introducing randomness.

Because of the considerable computational demands, only a restricted set of hyperparameters were investigated in the process of this study. Generally, all of the models (kNN, RF, XGBOOST) are trained via scikit-learn [24].

## 4. Results

The results of the training and evaluation process described in Section 3.3 are shown in Table 4.

**Table 4.** Model performance for binary and multiclass classification with  $n=8$  and  $n=100$  PCs.

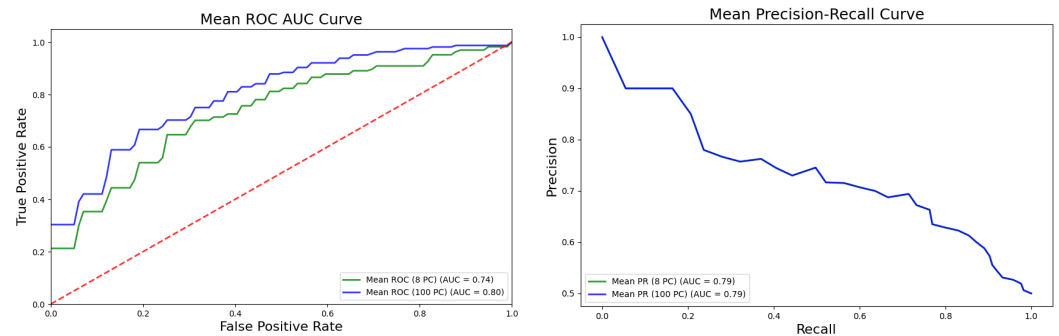
| Performance Metric                | RF    |        | kNN   |        | XGBoost |        |
|-----------------------------------|-------|--------|-------|--------|---------|--------|
|                                   | 8 PC  | 100 PC | 8 PC  | 100 PC | 8 PC    | 100 PC |
| <b>Multiclass Classification:</b> |       |        |       |        |         |        |
| Accuracy                          | 0.541 | 0.623  | 0.561 | 0.638  | 0.570   | 0.624  |
| ROC AUC                           | 0.855 | 0.839  | 0.754 | 0.817  | 0.770   | 0.846  |
| PR AUC                            | 0.557 | 0.668  | 0.567 | 0.647  | 0.562   | 0.673  |
| F1 Score                          | 0.530 | 0.610  | 0.538 | 0.603  | 0.554   | 0.606  |
| <b>Binary Classification:</b>     |       |        |       |        |         |        |
| Accuracy                          | 0.679 | 0.738  | 0.680 | 0.594  | 0.690   | 0.733  |
| ROC AUC                           | 0.743 | 0.804  | 0.735 | 0.625  | 0.758   | 0.811  |
| PR AUC                            | 0.790 | 0.790  | 0.758 | 0.636  | 0.787   | 0.834  |
| F1 Score                          | 0.671 | 0.735  | 0.640 | 0.373  | 0.680   | 0.723  |

### 4.1. Model Performance with Varying PC

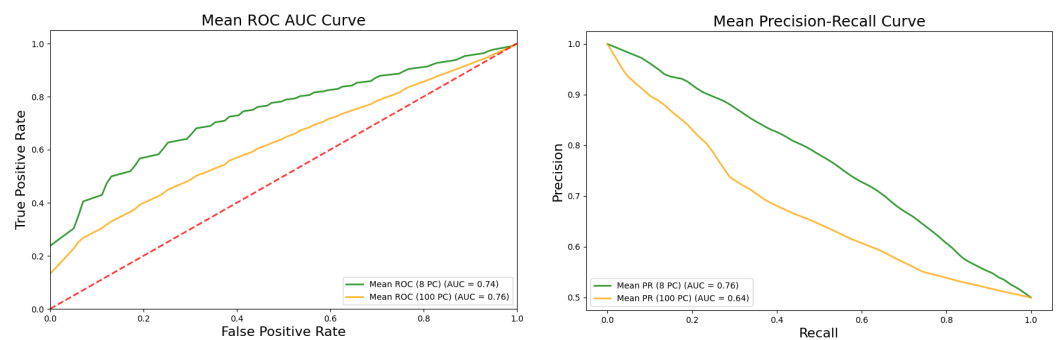
The performance metrics of the Machine Learning models—RF, kNN, and XGBoost—were evaluated for both binary and multiclass classification tasks using 8 PCs and 100 PCs (Table 4). The models exhibited notable differences in performance across these configurations.

This analysis highlights the significant impact of using different numbers of interaction features from PCA on the performance of various Machine Learning models. By comparing algorithms such as RF, kNN, and XGBoost using 8 and 100 PCs, variations in model performance can be observed based on different levels of feature complexity. While RF and XGBoost demonstrate enhanced performance with 100 PCs compared to 8 PCs, showing improvements in accuracy, ROC AUC, and F1 scores, kNN exhibits a different pattern. For multiclass classification, kNN shows modest improvement with increased PCs. However, for binary classification, kNN's performance declines across all metrics when using 100 PCs compared to 8 PCs.

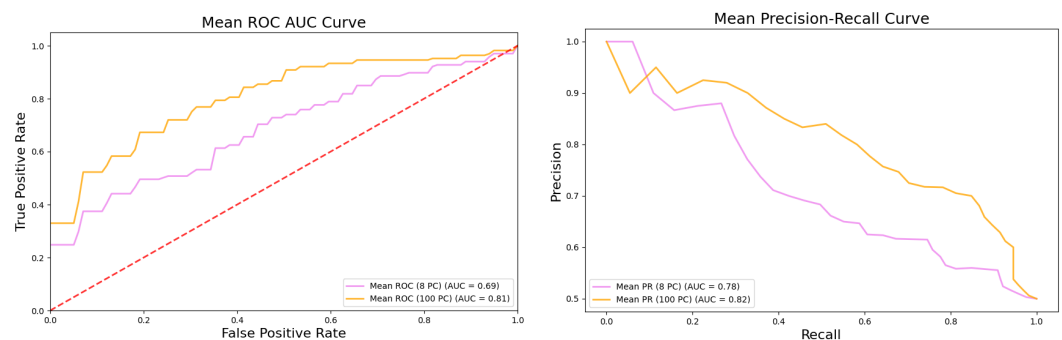
A visual representation of each model's ability (ROC AUC and Precision–Recall Curve) to discriminate between classes (binary classification) with different feature sets, and therefore increased dimensionality, are shown in Figures 4–6. Note: The red dashed line in the ROC AUC plots represents a random classifier, which acts as a baseline with an Area Under the Curve (AUC) of 0.5. This line signifies an equal increase in the True Positive Rate (TPR) and the False Positive Rate (FPR). Any ROC curve positioned above this dashed line indicates that the model's performance is better than random chance.



**Figure 4.** Performance Curves for RF: ROC AUC (left), PR AUC (right). Note: The PR AUC curves (right) for RF at 8 PCs and 100 PCs are identical (AUC = 0.79).



**Figure 5.** Performance Curves for kNN: ROC AUC (left), PR AUC (right). Note: The convex shape of the Precision–Recall curve for 100 PCs may result from retaining too many principal components, which reduces class separability and introduces noise—leading to less confident predictions and a rapid drop in precision as recall increases.



**Figure 6.** Performance curves for XGBoost: ROC AUC (left), PR AUC (right).

#### 4.2. Model Performance with Varying Classification Tasks (8 PCs)

Additionally, the performance of the RF, kNN, and XGBoost models was evaluated for both binary and multiclass classification tasks, using eight PCs that capture the base threshold of 95% of the variance in the data.

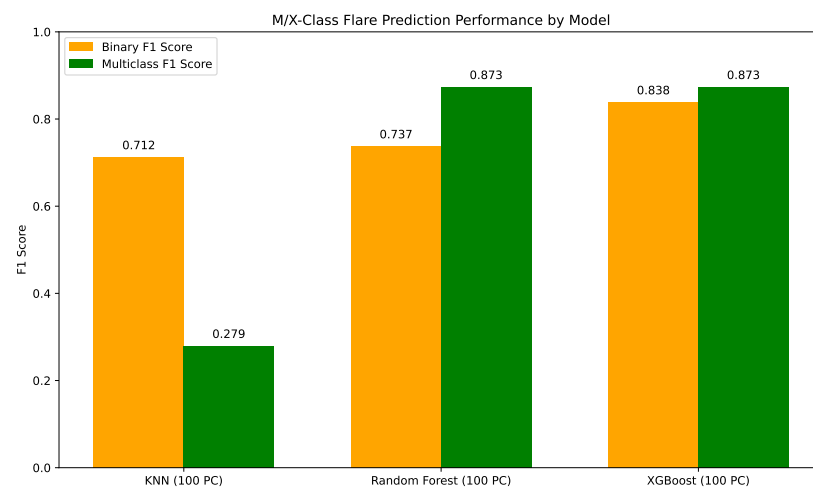
In binary classification, with eight PCs, the RF and XGBoost models demonstrated strong performance across metrics such as accuracy, ROC AUC, and F1 score. These models effectively captured the essential information from the reduced feature set, allowing them to make accurate predictions. On the other hand, the kNN model also performed reasonably well but showed slightly lower metrics compared to RF and XGBoost, indicating a modest ability to leverage the reduced dimensionality for binary classification.

For multiclass classification, the performance of the models with eight principal components varied. All metrics show a general decline across models, reflecting the increased difficulty of distinguishing between multiple flare classes. While XGBoost maintains relatively strong performance, kNN and RF struggle to achieve consistent scores across metrics. This suggests that with only 8 PCs, the feature space may be insufficiently expressive for reliable multiclass separation, highlighting the need for either more informative features or tailored model tuning.

Overall, with eight PCs, RF and XGBoost consistently outperformed kNN in both binary and multiclass classification tasks.

#### 4.3. Evaluation of Dangerous Flare Detection (M-/X-Class)

Figure 7 illustrates the comparative performance of the three models, kNN, RF, and XGBoost, in predicting M-/X-class solar flares. The F1 scores are shown for both BC and MC classification settings for 100 PCs. Across settings, the combined performance on M-/X-class flare prediction highlights XGBoost as the most effective model. These results indicate that XGBoost reliably detects dangerous flare events overall, though its MC performance is primarily driven by strong M-class detection. The consistently high combined scores suggest that, despite challenges in identifying rare X-class flares individually, XGBoost offers the most robust general performance for dangerous flare forecasting. For a more detailed breakdown of model performance across binary and multiclass settings, including weighted metrics and flare-specific sensitivity, see Appendix A.

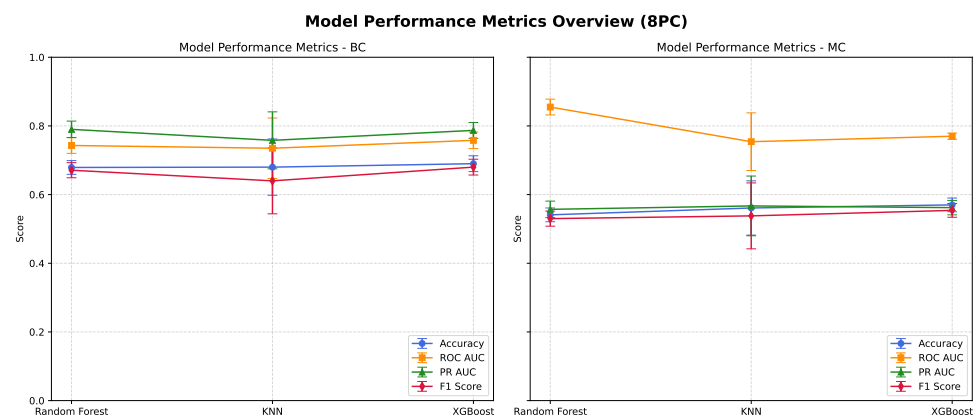


**Figure 7.** Comparison of F1 scores for M-/X-class flare prediction across binary and multiclass classification settings. XGBoost consistently achieves the highest performance, especially in multiclass classification.

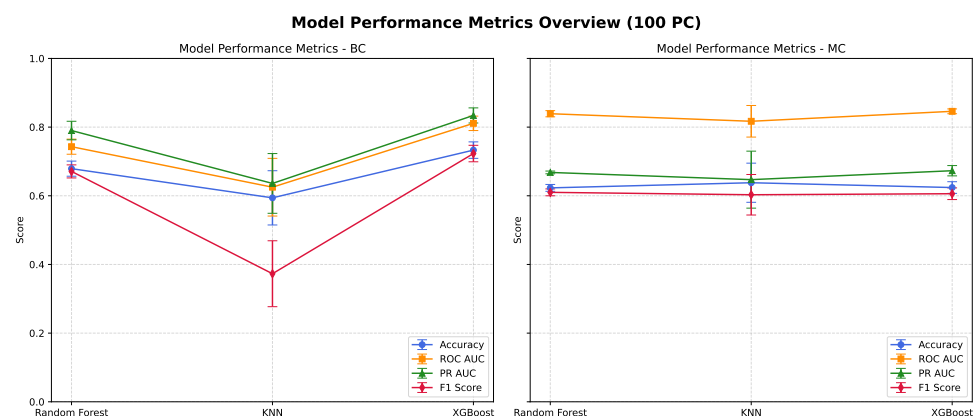
## 5. Discussion

The results from Table 4 regarding varying PCs suggest that RF and XGBoost effectively leverage the additional features provided by 100 PCs (capturing 97.5% of the variance of the original data set) to capture complex patterns and interactions in the data, resulting in better overall performance compared to the 8 PCs approach (capturing 95% of data set variance). The improvements in accuracy, ROC AUC, and F1 scores indicate that these models benefit from the increased dimensionality, or amount of input variables, allowing them to make more accurate predictions. The performance metrics for both models are competitive, with neither consistently outperforming the other across all metrics or classification tasks.

In contrast, kNN's performance behaviour indicates a sensitivity to the increased number of features, highlighted in Figures 8 and 9. This is a phenomenon often referred to as the 'curse of dimensionality', which refers to a Machine Learning model's performance deteriorating as the number of features (dimensions) or input variables increases.



**Figure 8.** Model performance Comparison across varying classifications (8 PCs). Note: MC: multiclass classification; BC: binary classification.



**Figure 9.** Model performance comparison across varying classifications (100 PCs). Note: MC: multiclass classification; BC: binary classification.

While it shows some improvement in multiclass classification with more PCs, the sharp decline in binary classification performance suggests that kNN may struggle with the higher dimensionality. This may be the case as kNN is an algorithm relying on the concept of distance to find its nearest neighbors. In high-dimensional spaces, the concept of proximity loses its meaning as data points become increasingly sparse. This makes it difficult for the algorithm to find truly relevant nearest neighbors, potentially leading to overfitting or difficulty in managing the increased complexity of the data.

These findings underscore the importance of carefully selecting the number of interaction features derived from PCA to optimize each model's performance.

The analysis of varying classification (binary and multiclass) of solar flare classes across models highlights key insights into the performance of RF, kNN, and XGBoost models using 8 PCs (PCs) for both binary and multiclass classification tasks, visualized in Figure 7. RF and XGBoost exhibited solid performance in both classification tasks, maintaining high accuracy, ROC AUC, and F1 scores, indicating their ability to effectively leverage the essential features captured by the principal components.

In contrast, the kNN model showed moderate performance in binary classification but a noticeable decline across all metrics in multiclass classification, where it lagged significantly behind RF and XGBoost.

Figure 8 further supports these findings by illustrating the comparative performance of the three models in predicting M/X-class solar flares using 100 PCs. XGBoost stands out with the highest combined F1 scores across both binary and multiclass settings, indicating its strong general capability in detecting dangerous flare events. However, this performance is largely driven by accurate M-class detection, while recall and F1 scores for X-class flares remain low. This highlights a common challenge in flare forecasting: identifying rare but critical events within imbalanced datasets.

Taken together, these results underscore the importance of selecting models that can handle both high-dimensional input and class imbalance effectively. While RF and XGBoost are well-suited for tasks involving larger feature sets, additional strategies, such as cost-sensitive learning or data augmentation, may be necessary to improve sensitivity to rare flare classes. kNN, though conceptually simple, may require dimensionality reduction or alternative distance metrics to remain competitive in such settings.

## 6. Conclusions

This study evaluated the predictive performance of Random Forest (RF), k-Nearest Neighbors (kNN), and Extreme Gradient Boosting (XGBoost) for solar flare classification using Principal Component Analysis (PCA). Models were tested on both binary and multiclass classification tasks using 8 and 100 Principal Components (PCs), capturing 95% and 97.5% of the dataset variance, respectively. The results show that RF and XGBoost consistently outperform kNN across most metrics, particularly when using 100 PCs. These tree-based models benefit from the increased dimensionality, which allows them to capture more complex patterns and interactions in the data. kNN, on the other hand, shows signs of performance degradation in binary classification with higher dimensionality, likely due to the curse of dimensionality affecting its distance-based approach. Further analysis of M/X-class flare prediction highlights XGBoost as the most effective model overall. It achieves the highest combined F1 scores in both binary and multiclass settings, indicating strong general performance in detecting dangerous flare events. However, its multiclass performance is primarily driven by accurate M-class detection, with limited sensitivity to rare X-class flares. This underscores the challenge of class imbalance and the need for targeted strategies to improve detection of critical but infrequent events. These findings emphasize the importance of selecting models that can handle both high-dimensional input and imbalanced data effectively. RF and XGBoost are well-suited for operational flare forecasting and other classification tasks requiring dimensionality reduction.

The insights gained from this study can significantly enhance future scientific research by improving solar flare prediction models, optimizing dimensionality reduction techniques, and informing model selection for astrophysical tasks.



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## Abbreviations

The following abbreviations are used in this manuscript:

|         |                                                    |
|---------|----------------------------------------------------|
| BC      | Binary Classification                              |
| MC      | Multiclass Classification                          |
| PCA     | Principal Component Analysis                       |
| PC      | Principal Components                               |
| ROC AUC | Receiver Operating Characteristic Area Under Curve |
| PR AUC  | Precision-Recall Area Under the Curve              |
| RF      | Random Forest                                      |
| XGBoost | Extreme Gradient Boosting                          |

## Appendix A. M/X-Class Flare Prediction Performance

For completeness, Appendix A Tables A1 and A2 present a comparative evaluation of M/X-class flare prediction performance across binary and multiclass classification tasks. While not central to the scope of this study, these results offer insight into model sensitivity to high-impact flare events. Notably, XGBoost consistently demonstrates superior performance across both settings, particularly in recall and F1 score, suggesting its robustness in identifying rare but critical solar activity.

To streamline the analysis and avoid redundancy, model comparisons in this section focus exclusively on the 100-PC configuration. This dimensionality setting consistently yielded stronger performance across all classifiers in preliminary evaluations, particularly in M/X-class flare detection.

F1 score was chosen as the primary metric because it balances precision and recall, making it ideal for evaluating performance on imbalanced datasets like solar flare classifications. Unlike accuracy, it reflects how well models detect rare but critical events such as X-class flares. This ensures a fair comparison across models in both binary and multiclass settings.

Given the inherent class imbalance in solar flare datasets, where M-class flares are more frequent than X-class flares, unweighted metrics may disproportionately reflect model performance on the dominant class. To ensure a more representative evaluation of predictive capability across both flare types, weighted performance metrics were employed. These metrics account for the actual distribution of M- and X-class samples in the test set, providing a balanced assessment and a realistic view of how well each model performs across both types. This helps ensure that the evaluation reflects the model's ability to detect the most critical events, not just the most common ones.

**Table A1.** Binary classification performance for M/X-class prediction.

| Model                  | Precision | Recall | F1 Score | Accuracy |
|------------------------|-----------|--------|----------|----------|
| kNN (100 PC)           | 0.778     | 0.667  | 0.712    | 0.667    |
| Random Forest (100 PC) | 0.714     | 0.770  | 0.737    | 0.727    |
| XGBoost (100 PC)       | 1.000     | 0.727  | 0.838    | 0.727    |

**Table A2.** Multiclass classification performance for M/X-class prediction (weighted by sample count: 142 M-Class; 23 X-Class).

| Model                  | Weighted Precision | Weighted Recall | Weighted F1 Score | Weighted Accuracy |
|------------------------|--------------------|-----------------|-------------------|-------------------|
| kNN (100 PC)           | 0.367              | 0.231           | 0.281             | 0.569             |
| Random Forest (100 PC) | 0.873              | 0.610           | 0.709             | 0.610             |
| XGBoost (100 PC)       | 0.891              | 0.856           | 0.875             | 0.856             |

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